WOLF QUANT MODEL

# Investment committee briefing

A disciplined global-equity decision system with portfolio constraints, stress testing and governance.

**DECISION: APPROVE A CONTROLLED LIVE PILOT**

| 55,504 | 60 | 87.5 / 100 | 0 |
|---|---|---|---|
| active equities | monthly decisions | governance score | hard breaches |

As of 13 August 2026 | CONDITIONALLY_APPROVED | Research evidence

# The decision in one page

The controls improved materially; the alpha claim did not.

## Recommendation

Approve a small, human-supervised live pilot. Do not authorize unattended or full-scale deployment yet.

## What still blocks full deployment

Observed PIT evidence is incomplete, and Wolf lagged equal weight by 1.06% per year in this 60-month sample (p=0.734).

## Why it earns a pilot

- Repeatable screening across six equity regions
- Adaptive VaR passes overall and chronological holdout tests
- Turnover and modeled cost drag now meet both 1.5 targets
- Zero hard breaches with auditable forecasts and trades

| Decision area | Evidence | Action |
|---|---|---|
| Process | Strong | Use |
| Risk backtest | 15 / 15 | Pass |
| Cost controls | Both targets met | Pass |
| Hard constraints | 0 breaches | Pass |
| PIT evidence | 7.5 / 15 | Improve |
| Deployable alpha | Not established | Do not claim |

**Approval means: Pilot capital, pre-trade review, monthly risk reporting and explicit stop conditions. It does not mean automatic execution.**

# What the model actually does

A monthly decision pipeline with risk controls at the end.

**1 OBSERVE**

Prices, filings, macro and portfolio

**2 FEATURE**

Point-in-time quality, value and risk signals

**3 FORECAST**

3, 6, 9 and 12 month distributions

**4 COMPARE**

Portfolio-aware, clean-sheet and LLM branches

**5 CONSTRAIN**

CVaR, liquidity and concentration limits

**6 GOVERN**

Stress, validation and committee reporting

## The output is a decision package

Ranked equities, target weights, trade direction, exposure limits, risk tests, stress results and an approval status.

## DRL is a challenger, not the final authority

Five real PPO seeds failed the validation hurdle and did not improve untouched-test Sharpe. The CVaR baseline remains at 100%.

**Human review stays between model recommendation and execution.**

# Evidence breadth and point-in-time progress



Delistings improved; observed historical availability remains the main gap.

| | | | |
|---|---|---|---|
| 55,504 | 1,409 | 156,960 | 59,183 |
| active equities | walk-forward eligible | historical forecasts | delistings archived |

| Region | Active | Master | Active share |
|---|---|---|---|
| DACH | 21,603 | 38,792 | 55.7% |
| EU ex-DACH | 2,436 | 5,088 | 47.9% |
| Hong Kong | 3,504 | 4,130 | 84.8% |
| Mainland China | 5,103 | 5,373 | 95.0% |
| UK | 3,932 | 6,664 | 59.0% |
| US | 18,926 | 52,523 | 36.0% |

| PIT evidence | Coverage | Status |
|---|---|---|
| Delisting events | 59,183 | Archived |
| Fundamentals with filing date | 7,081 | Mostly reconstructed |
| Observed acceptance times | 0 | Missing |
| Dated membership events | 0 | Missing |
| Inactive names with prices | 0 | Missing |
| Names with historical volume | 0 | Missing |

**Provider outcome: EODHD added 59,183 delistings; Nasdaq entitlement yielded five usable rows; Beam was unavailable; SEC blocked this runner. Unavailable history is never treated as observed.**

# Equities to establish in the target portfolio



Model trade direction versus current holdings; weights are targets.

| | |
|---|---|
| 19 | buy actions |
| 1 | reduce actions |
| 5.0% | target per name |
| 20 | target holdings |
| 6 | equity regions |

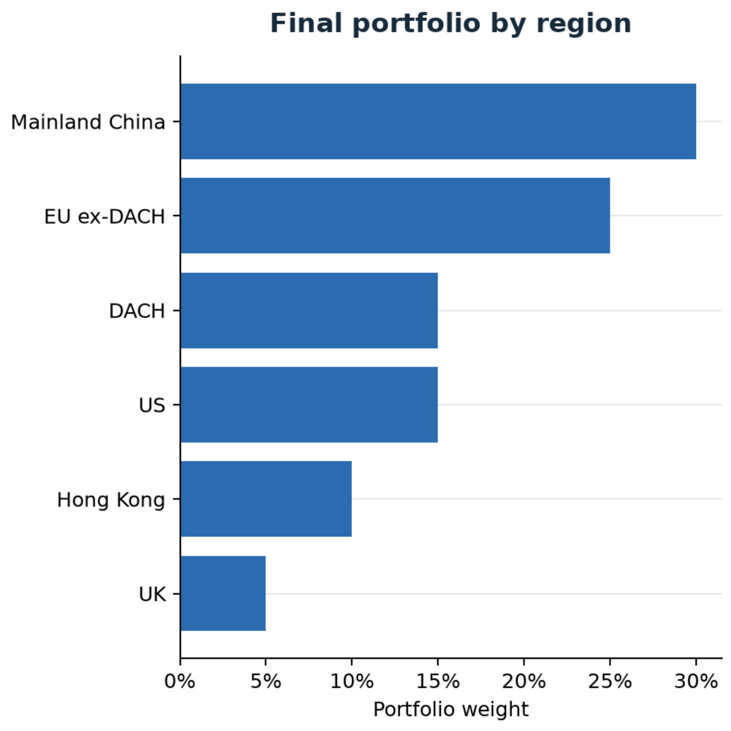
| Ticker | Company | Region | Action | Target |
|---|---|---|---|---|
| 000333.SHE | Midea Group Co Ltd | Mainland China | Buy | 5.0% |
| 000538.SHE | Yunnan Baiyao Group Co Ltd | Mainland China | Buy | 5.0% |
| 0823.HK | Link Real Estate Investment T | Hong Kong | Buy | 5.0% |
| 3988.HK | Bank of China Ltd H | Hong Kong | Buy | 5.0% |
| 600018.SHG | Shanghai International Port G | Mainland China | Buy | 5.0% |
| 600036.SHG | China Merchants Bank Co Ltd | Mainland China | Buy | 5.0% |
| 601816.SHG | Beijing-Shanghai High Speed R | Mainland China | Buy | 5.0% |
| 601818.SHG | China Everbright Bank Co Ltd | Mainland China | Buy | 5.0% |
| AD.AS | Koninklijke Ahold Delhaize NV | EU ex-DACH | Buy | 5.0% |
| ALV.XETRA | Allianz SE VNA O.N. | DACH | Buy | 5.0% |

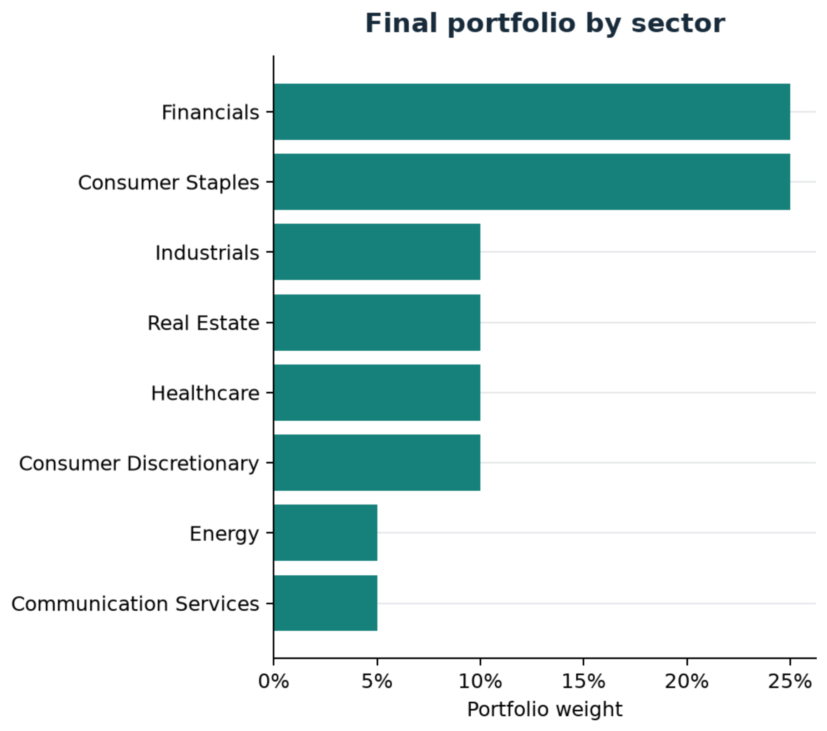
| Ticker | Company | Region | Action | Target |
|---|---|---|---|---|
| CS.PA | AXA SA | EU ex-DACH | Buy | 5.0% |
| ESSITY-B.ST | Essity AB (publ) | EU ex-DACH | Buy | 5.0% |
| HEN3.XETRA | Henkel AG & Co. KGaA vz. (Pre | DACH | Buy | 5.0% |
| LI.PA | Klepierre SA | EU ex-DACH | Buy | 5.0% |
| MCD.US | McDonalds Corporation | US | Buy | 5.0% |
| ORA.PA | Orange S.A. | EU ex-DACH | Buy | 5.0% |
| PG.US | Procter & Gamble Company | US | Buy | 5.0% |
| SBRY.LSE | J Sainsbury PLC | UK | Buy | 5.0% |
| SHEL.US | Shell PLC ADR | US | Buy | 5.0% |
| NOVN.SW | Novartis AG | DACH | Reduce | 5.0% |

**Execution note: recalculate orders using live NAV, FX, liquidity and prices. Manual review remains mandatory.**

# A deliberately diversified target

The portfolio uses equal name weights and explicit exposure caps.




## Construction rules

- 20 holdings at 5% each
- No sector above 25%
- No country above 30%
- No region or currency above 40%
- One listing per issuer
- No quarantined price history

### Largest allocations

Mainland China 30%; EU ex-DACH 25%; Financials and Consumer Staples 25% each.

# Why the CVaR portfolio is the baseline

It balances expected return, downside loss and diversification.

| Portfolio | Volatility | CVaR 5% | Yield | Hard breaches |
|---|---|---|---|---|
| Equal weight | 14.7% | -27.4% | 3.2% | 0 |
| Score weighted | 12.5% | -9.5% | 4.4% | 0 |
| Risk parity | 10.9% | -9.8% | 3.5% | 0 |
| Mean variance | 11.3% | -5.6% | 4.2% | 0 |
| CVaR constrained | 11.1% | -5.1% | 4.3% | 0 |
| Dividend income | 13.6% | -13.5% | 5.0% | 0 |
| Regime aware | 13.0% | -11.3% | 4.7% | 0 |

### Selected: CVaR constrained

The baseline keeps the forecast return target while reducing the average loss in the worst 5% of modeled outcomes to 5.10%.

## Why this matters

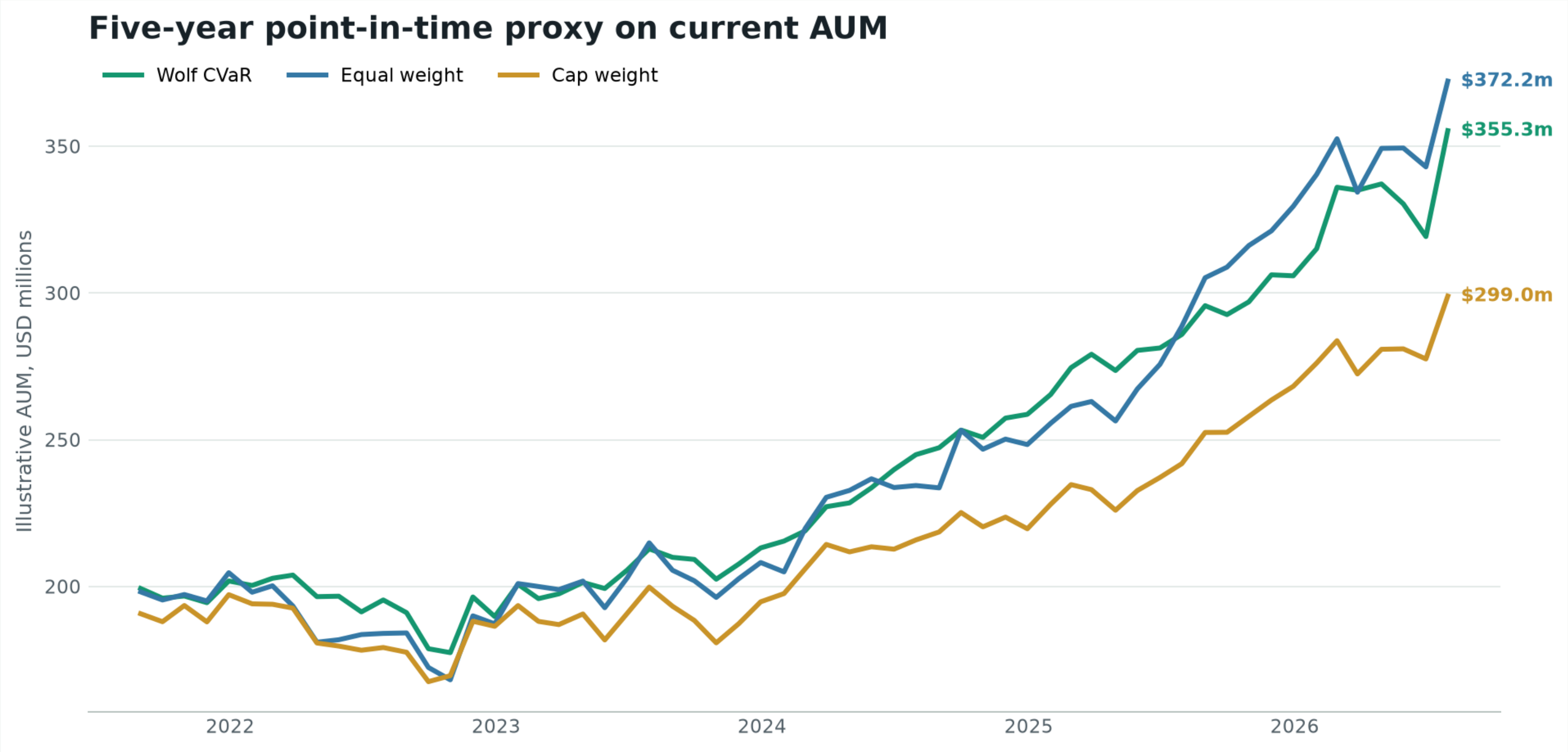
- Focuses on severe downside, not only average volatility
- Maintains 20 effective holdings
- Passes every hard portfolio constraint
- Produces the final 20-name equal-weight target

**Interpret carefully: Expected-return inputs are model forecasts, not guaranteed market returns. Live sizing still depends on trading capacity.**

# Five-year point-in-time proxy



The relevant model evidence uses dated monthly decisions.




## Illustrative current-AUM result

$186.1m grows to $355.3m; net PnL is $169.3m after trading costs and before the bank fee.

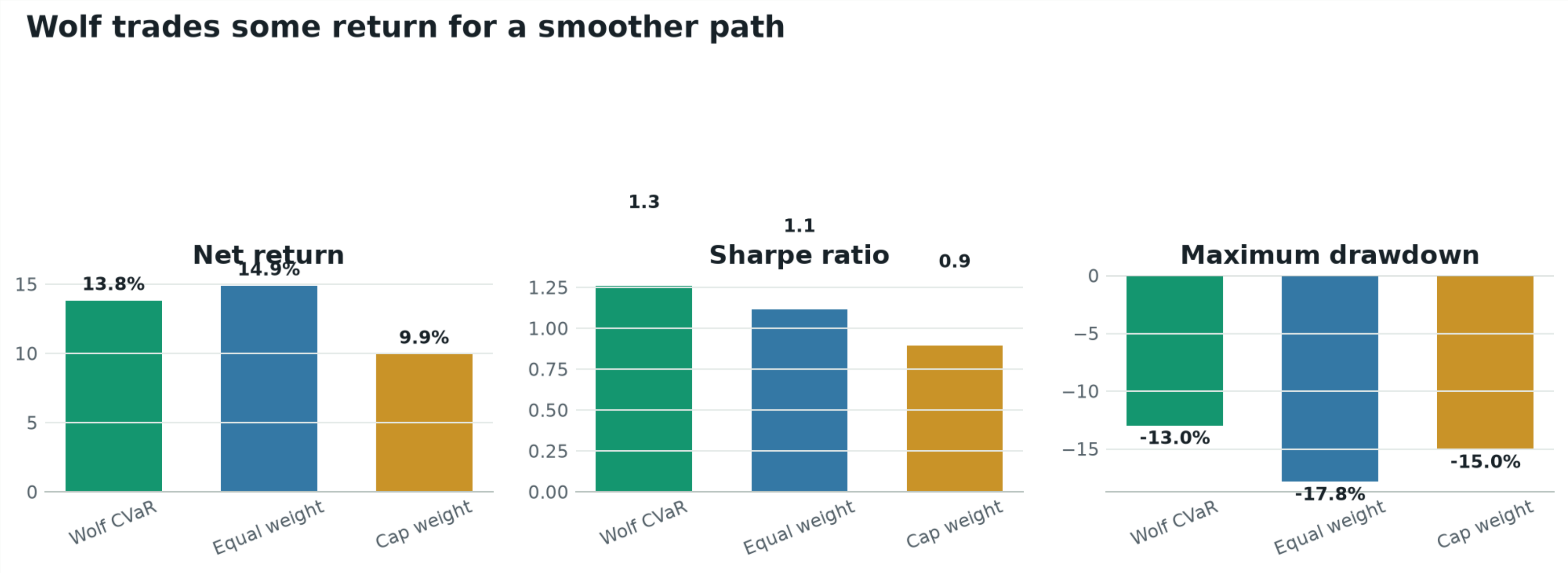
- Net annualised return: 13.8%
- Sharpe ratio: 1.26
- Maximum drawdown: -13.0%
- Positive months: 65.0%

## Not a forecast

This scales the realised proxy path to current AUM. The 25 bp annual bank fee is modeled separately in the long replay.

# What beat what over 60 months

Wolf delivered the best risk-adjusted result, not the best return.




## The useful result

Wolf Sharpe was 1.26 and drawdown was 13.0%, better than both simple controls on these risk measures.

## The honest result

Equal weight returned 14.88% versus Wolf at 13.81%. The model did not beat the simplest control on net return.

## Investment interpretation

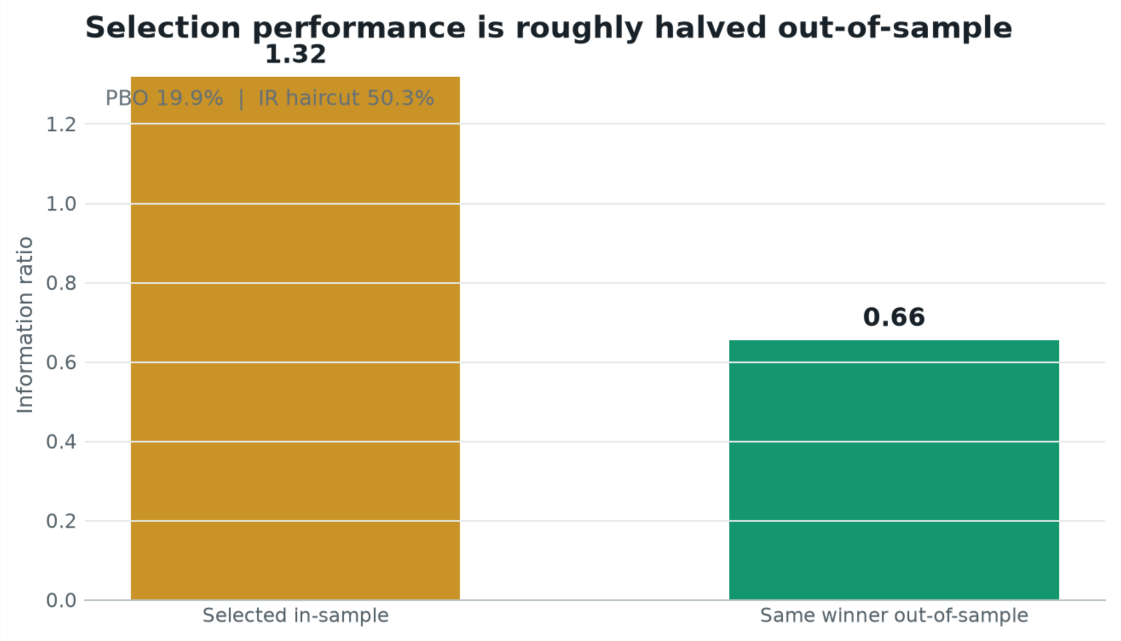
The current case is risk control and decision discipline. Incremental stock-selection alpha remains unproven.

**Highest return: Equal weight | Highest Sharpe: Wolf | Smallest drawdown: Wolf**

# Alpha and overfitting: the hard truth

The retrospective strategies are interesting; deployable alpha is not established.

| Benchmark | Active return | Reg. alpha | p-value | Verdict |
|---|---|---|---|---|
| Cap weight | 3.43% | 4.40% | 0.108 | NOT SIGNIFICANT |
| Equal weight | -1.24% | 3.29% | 0.312 | NOT SIGNIFICANT |




## What the 50.3% fall means

When history selects the apparent winner, its information ratio typically drops from 1.32 in-sample to 0.66 out-of-sample. About half the apparent edge does not survive selection.

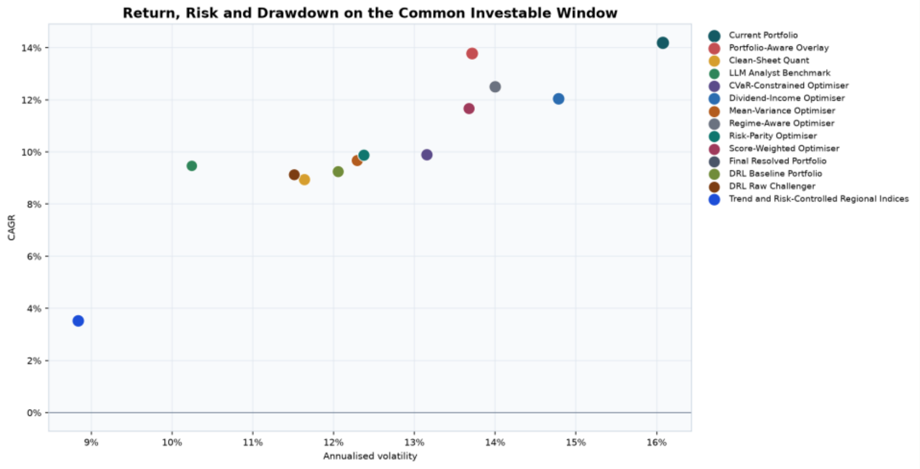
## Decision

Use Wolf for disciplined ranking and risk control. Do not market it as a proven alpha engine until native live evidence clears the benchmark and cost hurdles.

# The 1997 replay is a stress map

It shows how today's holdings behaved through history, not what the model knew then.

| Portfolio | Start | CAGR | Sharpe | Max DD | End |
|---|---|---|---|---|---|
| Current Portfolio | $186.1m | 14.1% | 0.75 | -42.6% | $9.0bn |
| Portfolio-Aware Overlay | $186.1m | 13.0% | 0.77 | -41.2% | $6.8bn |
| Risk-Parity Optimiser | $100.0k | 10.9% | 0.77 | -30.6% | $2.1m |
| LLM Analyst Benchmark | $100.0k | 9.5% | 0.81 | -20.9% | $1.5m |
| Final Resolved Portfolio | $186.1m | 9.1% | 0.65 | -28.4% | $2.4bn |




## What looks strongest

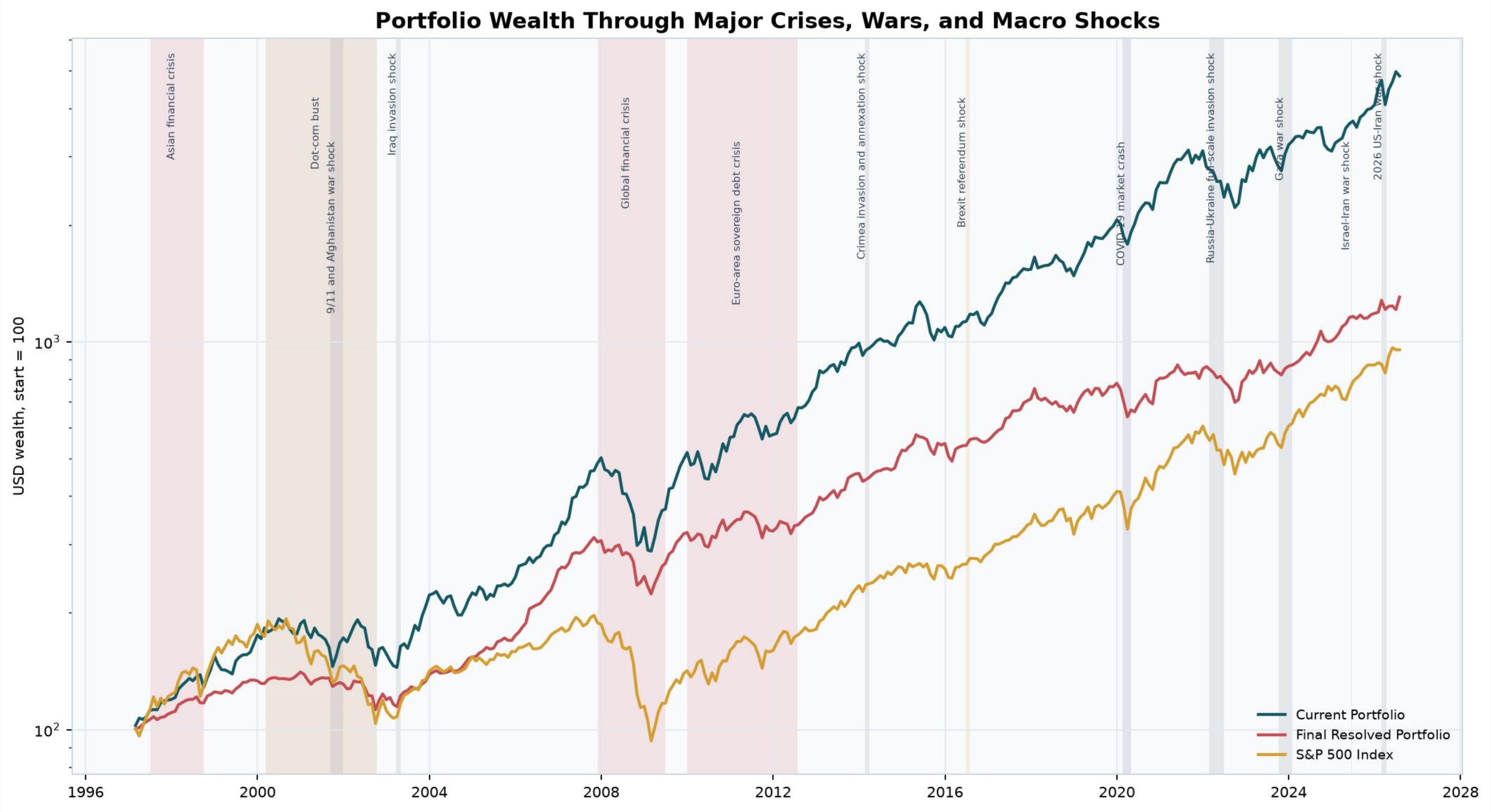
Current Portfolio PnL: $8.8bn. Final Resolved PnL: $2.2bn. Risk Parity led the independent optimisers.

## Why it cannot prove selection skill

The securities were chosen with current information and then replayed backward. Survivorship and selection look-ahead make the long PnL unsuitable as a live promise.

# Performance through major market shocks

Event windows are descriptive overlaps, not causal estimates.



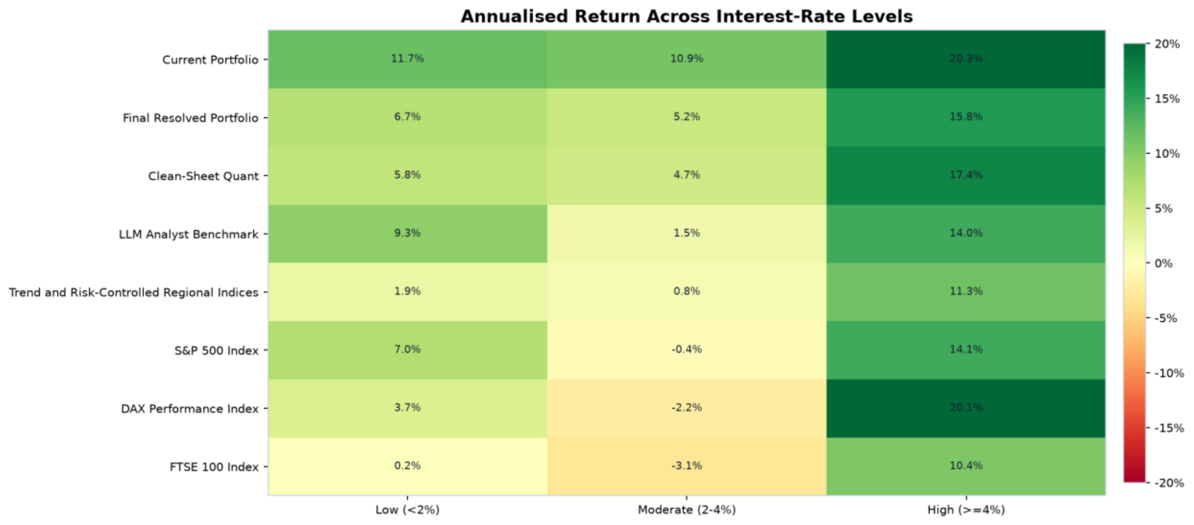

**COVID-19: -12.58% return; -15.19% event drawdown**

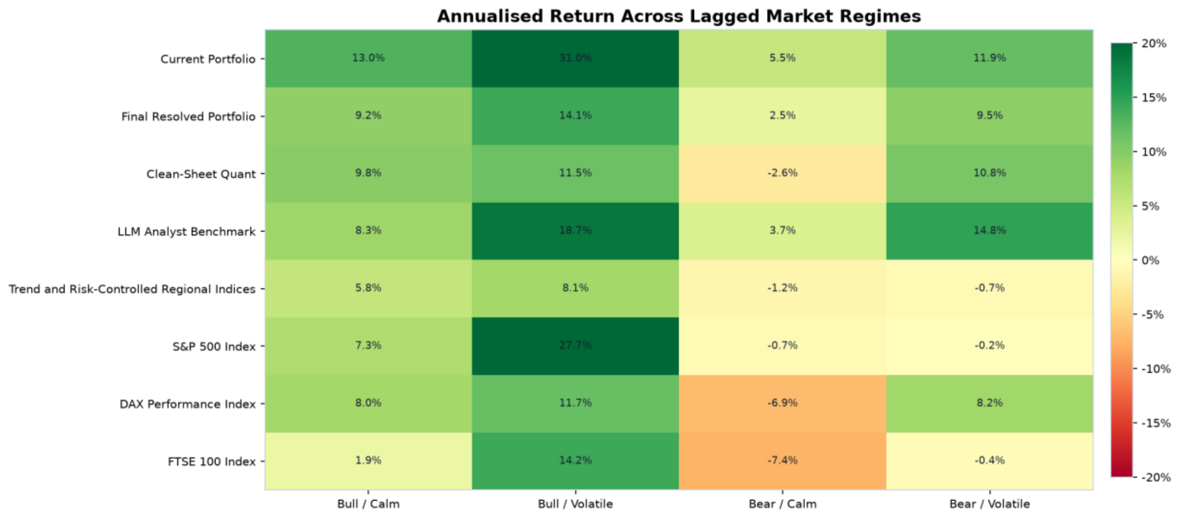
**2008 crisis: -27.12% event drawdown**

**2026 Iran war: +3.57% return; -5.27% drawdown**

# Interest-rate and market-regime context

Conditional groupings show where the final portfolio was most comfortable.



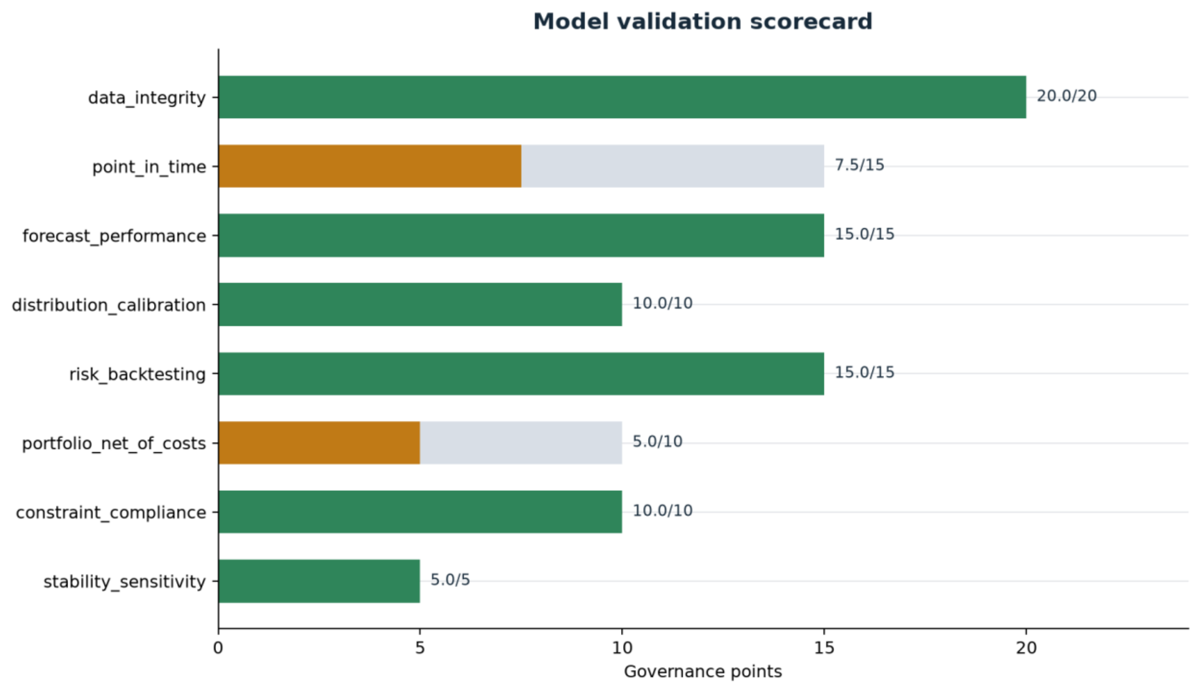




**Rate reading: Final Portfolio was strongest in high-rate months (15.77%) and when rates were rising (13.03%).**

**Regime reading: Its strongest market group was Bull / Volatile (14.09%). Expansion months returned 9.91% annualised.**

# Risk calibration now passes; approval stays conditional

Adaptive VaR fixed exception clustering without hiding PIT and alpha limits.




## VaR coverage and independence

| Sample | VaR | Exceptions | Kupiec p | Indep. p |
|---|---|---|---|---|
| Overall | 95% | 62/1152 | 0.557 | 0.367 |
| Overall | 99% | 10/1152 | 0.645 | 0.072 |
| Holdout | 95% | 21/460 | 0.664 | 0.074 |
| Holdout | 99% | 3/460 | 0.423 | 0.843 |

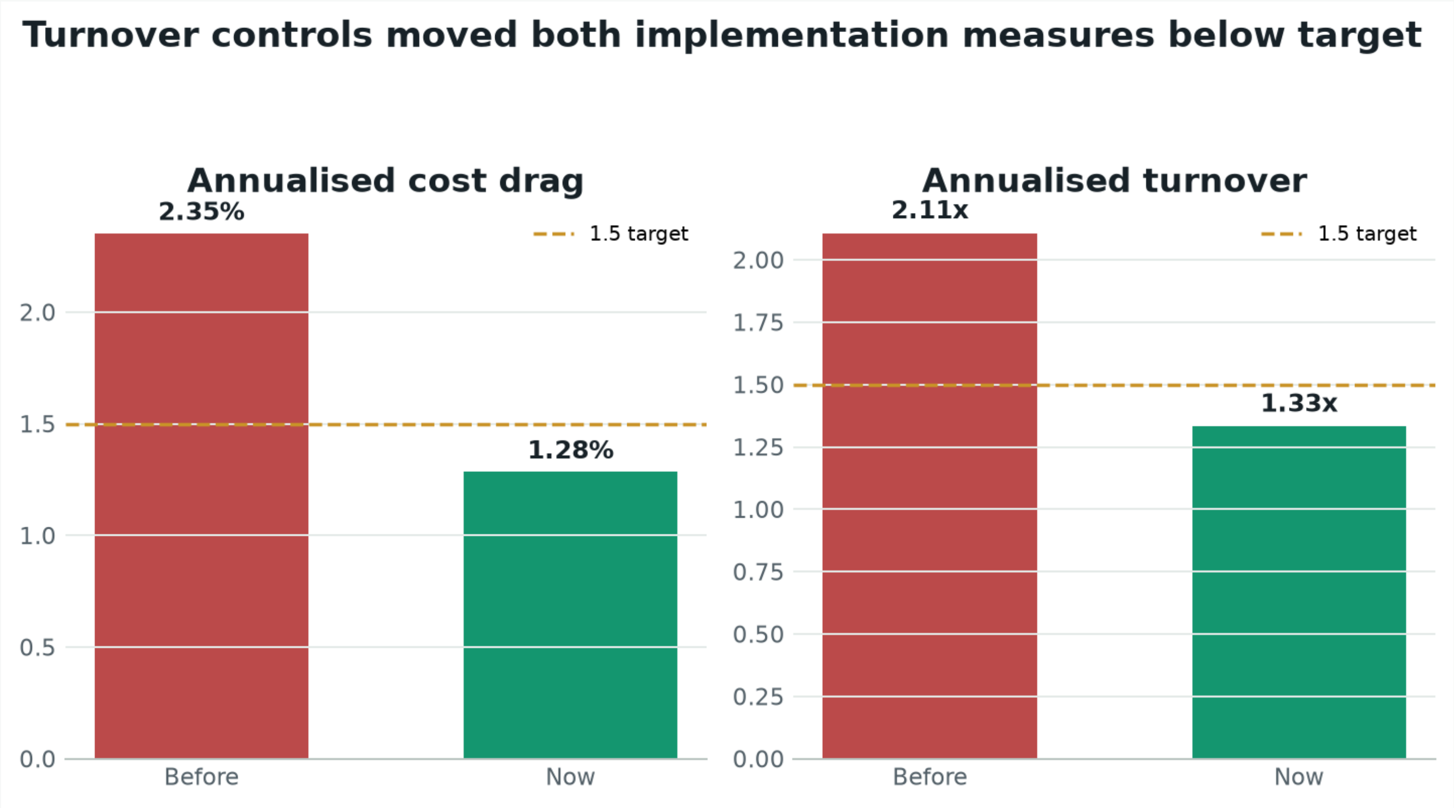
### Adaptive risk stack: 15 / 15

DCC-IGARCH Student-t, filtered historical simulation, EWMA Normal and EWMA Student-t are selected using trailing data. The holdout is chronological reconstructed evidence, not a pristine future shadow period.

**Overall 87.5/100 | 6 pass | PIT and portfolio warnings | 0 critical failures | full local suite passed**

# Turnover and modeled cost drag now meet target

Retention hysteresis and minimum-turnover transitions reduced churn.




**1.28%**

modeled cost drag; target <= 1.50%

**1.33x**

annual turnover; target <= 1.50x

**$465k**

0.25% annual bank fee at reference AUM

### Why portfolio remains a warning

Both cost gates pass, but Wolf trailed equal weight by 1.06% per year and the difference was not significant (p=0.734).

**Retention and capped/minimum-turnover transitions drove the improvement; the configured no-trade band did not trigger in this sample.**

# A controlled path to live use

Scale only when operations, costs and evidence earn it.

### 0 SHADOW

**4 weeks**

Generate orders, do not trade. Validate data, NAV, FX and fills.

### 1 PILOT

**6 to 12 months**

Use limited capital with human approval and staged execution.

### 2 SCALE

**Evidence gated**

Increase capital only after cost, risk and benchmark hurdles hold.

## Proposed pilot gates

- Zero hard constraint or data-lineage breaches
- Pre-trade cost estimate approved for every rebalance
- Keep turnover <=1.5x; stretch goal <=1.0x in the pilot
- Net performance tracked against equal and cap weight
- Monthly model-risk and investment-committee review

## Immediate stop conditions

Stale critical data, any hard breach, a failed live risk test, failed reconciliation or trading costs above the approved budget.

**Alpha gate: Full-scale alpha claims require native live evidence and at least 60 months under the existing governance policy.**

# How to read the results



Plain-language definitions for the investment committee.

| Measure | Simple meaning |
| --- | --- |
| CAGR | Average compounded annual growth. |
| Sharpe | Return earned per unit of overall variability. |
| Sortino | Return earned per unit of harmful downside variation. |
| Max drawdown | Largest peak-to-trough loss. |
| CVaR | Average loss among the worst modeled outcomes. |
| Information ratio | Active return per unit of benchmark-relative risk. |
| p-value | How surprising the result is if true alpha were zero. |
| PBO | Chance that selecting the backtest winner overfits history. |

## Three evidence layers

1. Dated 60-month proxy: primary model evidence.
2. 1997 replay: stress and exposure history.
3. Native live record: required before full alpha approval.

## Higher is not always better

A portfolio can have a higher return but a deeper drawdown, more turnover or weaker statistical evidence. The committee should judge the package, not one headline number.

## Where to verify

The GitHub release contains the 42-page PDF, plain-language interpretation, CSV evidence, plots, manifests and checksums.

THE RECOMMENDATION

# Approve a controlled live pilot.

Wolf is ready to improve how decisions are made: broad screening, explicit risk, consistent portfolio construction and an audit trail. It is not yet ready to promise alpha.

### USE NOW

Ranking, diversification, adaptive risk and governance

### PROVE LIVE

Net benchmark edge in a genuine future shadow record

### FIX NEXT

Observed PIT vintages and probability calibration

**Investment edge today: a better-controlled process. Investment edge tomorrow: only what live evidence proves.**